MY_PRINTF_

< PRINTF COMMAND-LIKE />

MY_PRINTF

binary name: libmy.a
language: C
compilation: via Makefile, including re, clean and fclean rules

- ✓ The totality of your source files, except all useless files (binary, temp files, obj files,...), must be included in your delivery.
- ✓ All the bonus files (including a potential specific Makefile) should be in a directory named bonus.
- ✓ Error messages have to be written on the error output.

You must recode the **printf** function from the C library according to the C99 standard. Your function should be prototyped as the printf function.
You do not have to implement the C library printf buffer handling.

man 3 printf / man 3 stdarg

You must process all **printf** flags.

You must submit a Makefile that will create a library named my, as well as all source files. The libmy.a library must contain the my_printf function, in addition to any other functions required to make it functional.

The whole libC is forbidden, except **va_start**, **va_end**, **va_arg**, **malloc**, **free** and **write**.

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Makefile

In your `Makefile` you must list all of the `c` files separatly (no `*.c`), compile them, and use the result to generate the `libmy.a` library.

Your `Makefile` must have the following rules:

- ✓ libmy.a
- ✓ all (which calls the libmy.a rule)
- ✓ clean
- ✓ fclean (which calls the clean rule)
- ✓ re (which calls the fclean and libmy.a rules)
- ✓ unit_tests (which calls the fclean and libmy.a rules, and then links the lib with the tests) - **Optional**
- ✓ tests_run (which calls the unit_tests rule and executes the unit_tests bin) - **Optional**

Unit tests

Criterion includes mechanisms to test standard output and standard error, you can learn more about it there...

```c
#include <criterion/criterion.h>
#include <criterion/redirect.h>
#include "my.h"

void redirect_all_std(void)
{
    cr_redirect_stdout();
    cr_redirect_stderr();
}

Test(my_printf, simple_string, .init = redirect_all_std)
{
    my_printf("hello world");
    cr_assert_stdout_eq_str("hello world");
}
```

Examples

```c
char str[8];

my_strcpy(str, "world !");
my_printf("Hello %s\n", str);
```

Terminal

```
$>  ./a.out | cat -e
Hello world !$
$>
```

Bonuses

Be creative, you might also look for every versions of printf like sprintf, fprintf, dprintf, etc...
%S and %b are considered as bonuses.

Every bonuses have to be in the bonus directory

v 1.2

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